

BARRIE, ON, CanadaWMO: **714361**Lat: **44.380N**Lon: **79.780W**Elev: **295**StdP: **97.83**Time Zone: **-5.00 (NAE)**Period: **94-03**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.3	-19.0	-28.3	0.3	-21.8	-24.5	0.4	-18.5	9.9	-6.6	9.0	-5.2	2.6	110	0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	29.9	22.1	28.0	21.3	26.3	20.0	23.3	28.1	22.2	26.7	21.1	24.9	3.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.8	17.0	26.2	20.6	15.9	24.5	19.7	14.9	23.3	70.9	28.4	66.4	26.5	62.3	24.9	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
8.5	7.5	6.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			-27.7	24.8	4.2	2.2	-30.8	26.4	-33.3	27.7	-35.6	28.9	-38.7	30.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-7.3	-6.0	-1.7	5.1	11.4	17.4	19.4	18.7	14.8	8.5	3.0	-3.8
	DBStd	10.64	6.86	6.01	6.10	5.08	4.55	4.24	3.17	3.06	4.20	4.38	4.68	5.88
	HDD10.0	2281	536	448	364	161	36	2	0	0	7	84	214	427
	HDD18.3	4408	794	682	619	397	219	67	25	33	121	306	461	685
	CDD10.0	1077	0	0	3	15	81	224	292	270	152	37	4	1
	CDD18.3	163	0	0	0	1	6	38	59	44	15	0	0	0
	CDH23.3	1304	0	0	0	13	56	337	449	324	119	5	0	0
	CDH26.7	311	0	0	0	2	6	99	104	76	23	0	0	0
Wind	WSAvg	2.9	3.4	3.4	3.3	3.5	3.1	2.5	2.2	1.9	2.3	2.7	3.3	3.2
Precipitation	PrecAvg	834	60	52	56	66	73	78	81	77	76	72	73	69
	PrecMax	1062	104	106	144	131	132	207	119	168	135	134	136	108
	PrecMin	623	28	26	18	22	26	21	33	35	30	23	25	30
	PrecStd	98	20	23	27	27	25	41	27	29	25	28	27	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	8.9	18.8	25.4	27.4	31.3	31.8	32.1	29.6	23.9	16.5	12.5
		MCWB	7.2	7.3	13.3	16.5	19.6	23.5	24.0	22.3	21.2	18.0	12.4	11.0
	2%	DB	4.8	6.3	13.8	20.3	24.9	29.6	29.4	28.5	26.5	20.6	14.2	8.6
		MCWB	3.8	4.2	10.2	14.9	18.0	22.1	22.3	21.7	19.9	16.0	12.2	7.3
	5%	DB	2.9	4.2	9.6	16.5	22.5	27.5	27.6	26.6	24.3	17.9	12.3	5.7
		MCWB	1.9	2.6	6.1	10.7	16.3	20.9	21.1	20.3	18.9	14.8	10.5	4.7
	10%	DB	1.2	1.9	6.3	13.4	19.7	24.9	25.8	24.9	21.8	15.6	10.1	3.4
		MCWB	0.5	0.6	4.1	8.3	14.4	19.3	19.7	19.7	17.4	12.8	8.6	2.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	7.1	15.0	18.6	20.7	24.5	24.9	24.1	22.8	18.5	14.2	11.1
		MCDB	7.6	8.0	18.0	23.3	25.5	30.2	29.9	28.6	26.5	22.3	15.6	12.0
	2%	WB	4.1	4.5	10.4	15.1	18.9	22.8	23.5	22.7	20.9	16.5	12.6	7.7
		MCDB	4.6	6.0	13.6	19.3	23.8	28.4	28.3	26.8	24.7	19.6	13.9	8.6
	5%	WB	2.0	2.8	6.6	11.7	17.1	21.5	22.0	21.5	19.5	15.0	10.6	4.8
		MCDB	2.7	4.3	9.1	15.6	21.6	26.3	26.3	25.3	22.8	17.6	11.8	5.6
	10%	WB	0.5	0.8	4.0	9.0	15.0	20.2	20.8	20.4	18.3	13.3	8.4	2.4
		MCDB	1.3	1.9	6.3	12.8	18.3	23.8	24.4	23.6	21.2	15.4	10.1	3.5
Mean Daily Temperature Range	5% DB	MDBR	7.6	8.2	8.5	9.9	10.8	10.4	10.6	10.7	10.4	8.9	6.8	6.7
		MCDBR	6.7	8.6	12.2	15.7	13.6	13.4	12.7	13.1	13.6	12.7	9.8	7.9
		MCWBR	6.3	7.5	9.0	10.5	8.6	7.4	6.7	6.8	7.9	8.7	8.3	6.7
	5% WB	MCDBR	6.5	8.4	12.0	14.0	12.5	11.9	11.3	10.8	11.8	11.8	9.1	7.1
		MCWBR	6.2	7.7	9.5	10.3	8.7	7.0	6.7	6.5	7.4	8.7	8.3	6.7
Clear-Sky Solar Irradiance	taub	0.280	0.295	0.315	0.351	0.382	0.402	0.399	0.391	0.358	0.340	0.307	0.280	
	taud	2.381	2.323	2.410	2.365	2.331	2.292	2.321	2.332	2.423	2.461	2.502	2.443	
	Ebn at Noon	846	894	924	912	888	868	866	860	865	834	812	813	
	Edh at Noon	75	96	101	115	123	129	124	118	100	84	68	65	
All-Sky Solar Radiation	RadAvg	1.36	2.23	3.54	4.49	5.39	5.89	5.95	5.12	4.04	2.36	1.40	0.99	
	RadStd	0.16	0.28	0.37	0.44	0.54	0.43	0.39	0.27	0.39	0.25	0.20	0.12	